

Prop. Let X be a Compact Hausdorff space. Then
 $A \subseteq X$ is Compact $\xleftrightarrow{T_2}$ if and only if $A \subseteq X$ is closed
 $\xleftarrow{\text{Compact.}}$

Proof.

Prop

Let X be any space and Y be Compact Hausdorff space.

$f: X \rightarrow Y$ is continuous map if and only if the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

Proof. Let $f: X \rightarrow Y$ be a continuous map.

Let $(x, y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets U, V

$$U \cap V = \emptyset, \quad y \in U, \quad f(x) \in V$$

Since $f(x) \in V \stackrel{\text{def}}{\iff} x \in f^{-1}[V]$ and f is continuous, $f^{-1}[V]$ is an open set in X .

Now, $(x, y) \in f^{-1}[V] \times U$

And, $(f^{-1}[V] \times U) \cap G_f = \emptyset$, because if $(x, f(x)) \in f^{-1}[V] \times U$, then

$f(x) \in V$ and $f(x) \in U$, it is contradiction to disjointness.

Therefore, $(x, y) \in f^{-1}[V] \times U \subseteq X \times Y \setminus G_f$, so proved. $\square$

Conversely, let G_f is closed in $X \times Y$, under the Product Topology.

Let $C \subseteq Y$ be a closed set in Y . Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $P_X: X \times Y \rightarrow X: (x, y) \mapsto x$ is closed map if Y is compact. $\square$

Note: for $g = f \times i: X \times Y \rightarrow Y \times Y: (x, y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x, y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f$$

Prop.

Compact Hausdorff space is Normal space.

Proof. Let disjoint closed subsets C and D be given.

Let $x \in C$ be fixed. For each $y \in D$, since $C \cap D = \emptyset$, $y \neq x$.

By the T_2 condition, there exists disjoint open sets V_y and U_y , containing x and y respectively. Then, $\{U_y \mid y \in D\}$ is open cover for D and since D is closed set in the compact space, so it is compact. Therefore, there is finite subcover $\{U_{y_i} \mid i \in I\}$, where I is finite indices set. Put

$$V_x = \bigcap_{i \in I} V_{y_i} \quad \text{and} \quad W_x = \bigcup_{i \in I} U_{y_i}.$$

Since I is finite, V_x is open set containing x and $V_x \cap U_{y_i} = \emptyset$ for all $y_i \in D$ implies $V_x \cap W_x = \emptyset$, and $D \subseteq W_x$.

Since $x \in C$ was arbitrary, we can construct an open cover for C ,

$$\{V_x \mid x \in C\}$$

Similarly, C is compact, so there is finite open cover

$$\{V_{x_j} \mid j \in J\}$$

where J is finite. Now,

$$C \subseteq \bigcup_{j \in J} V_{x_j} \quad \text{and} \quad D \subseteq \bigcap_{j \in J} W_{x_j} \quad (\text{It is also open since } J \text{ finite})$$

and V_{x_j} is disjoint for some W_{x_k} , always thus these open sets are disjoint. $\square$